

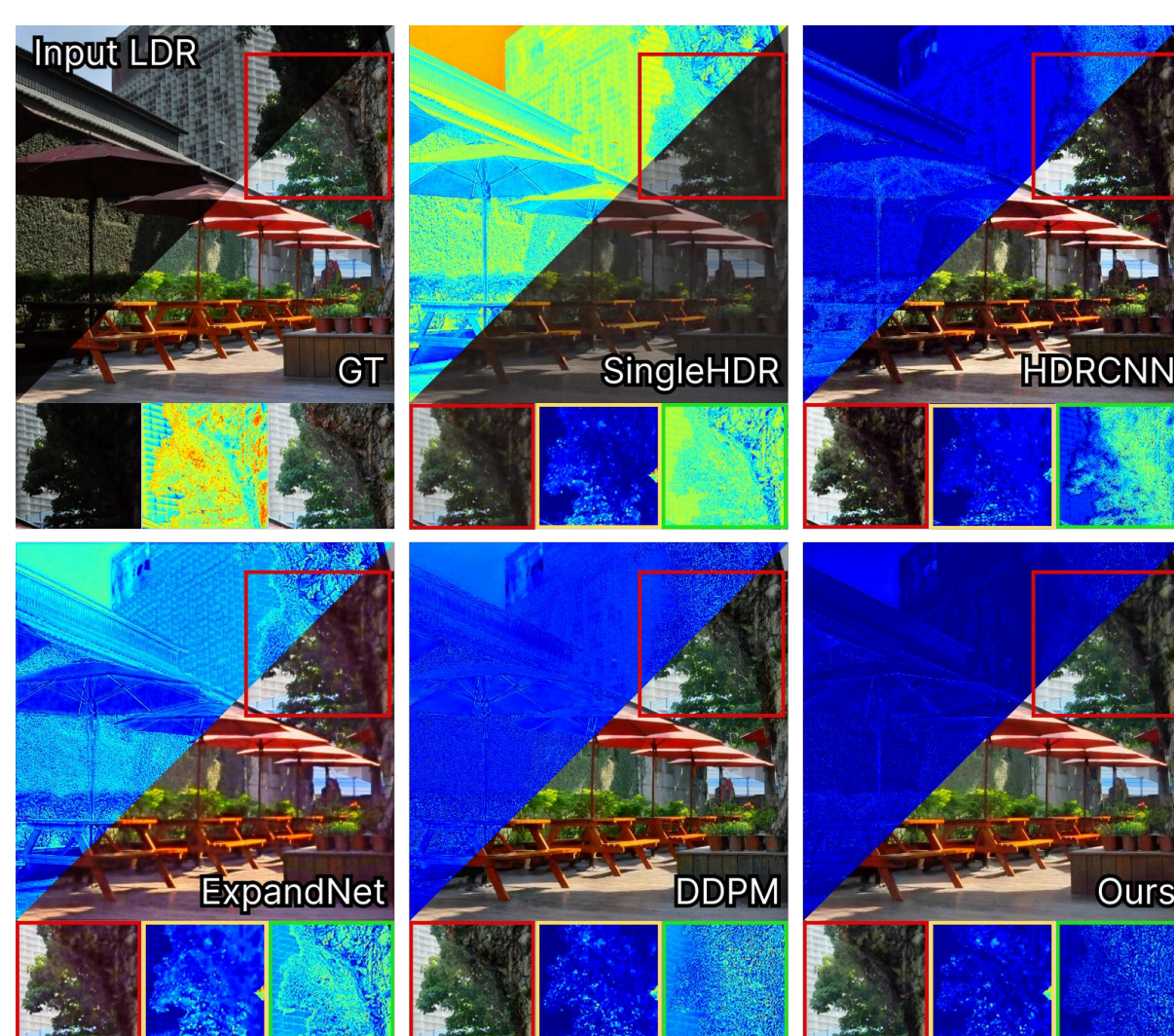
Motivation

Challenge:

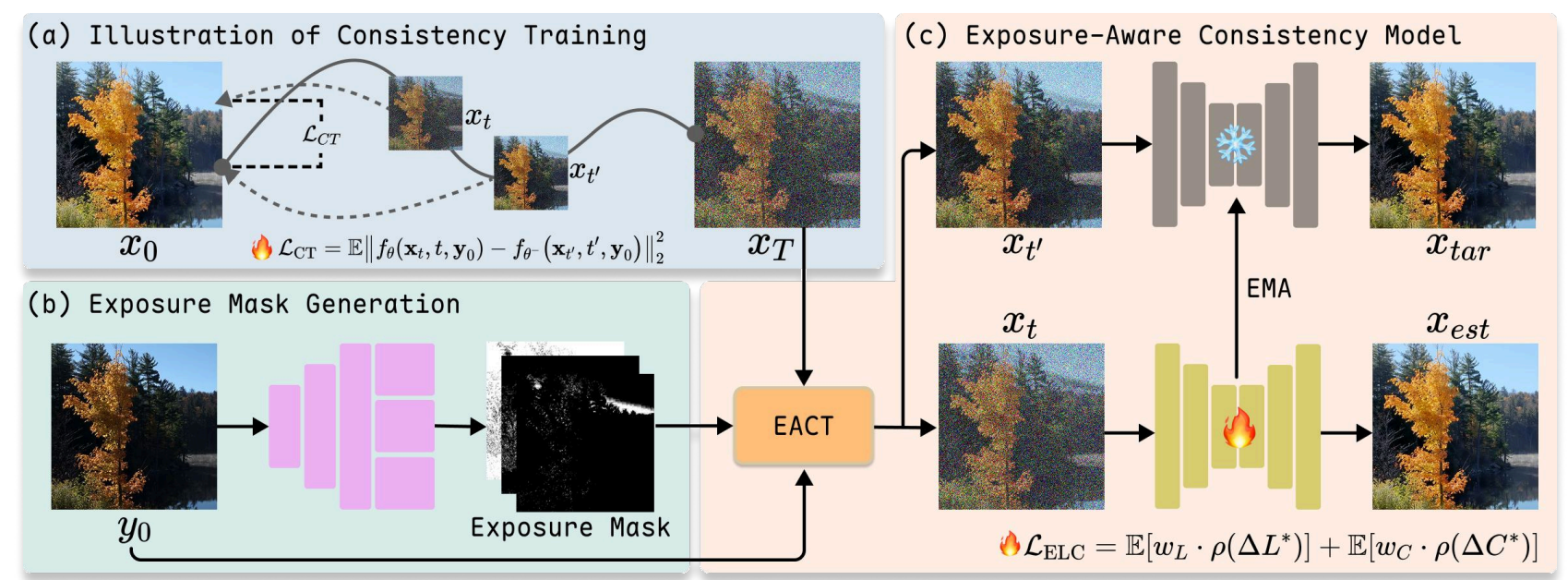
Single-image LDR-to-HDR reconstruction is highly ill-posed due to detail loss in over-exposed regions and noise amplification in under-exposed areas.

Current Methods:

1. Directly regress HDR from LDR inputs, often missing degraded details.
2. Use costly diffusion processes with limited exposure awareness.



Method



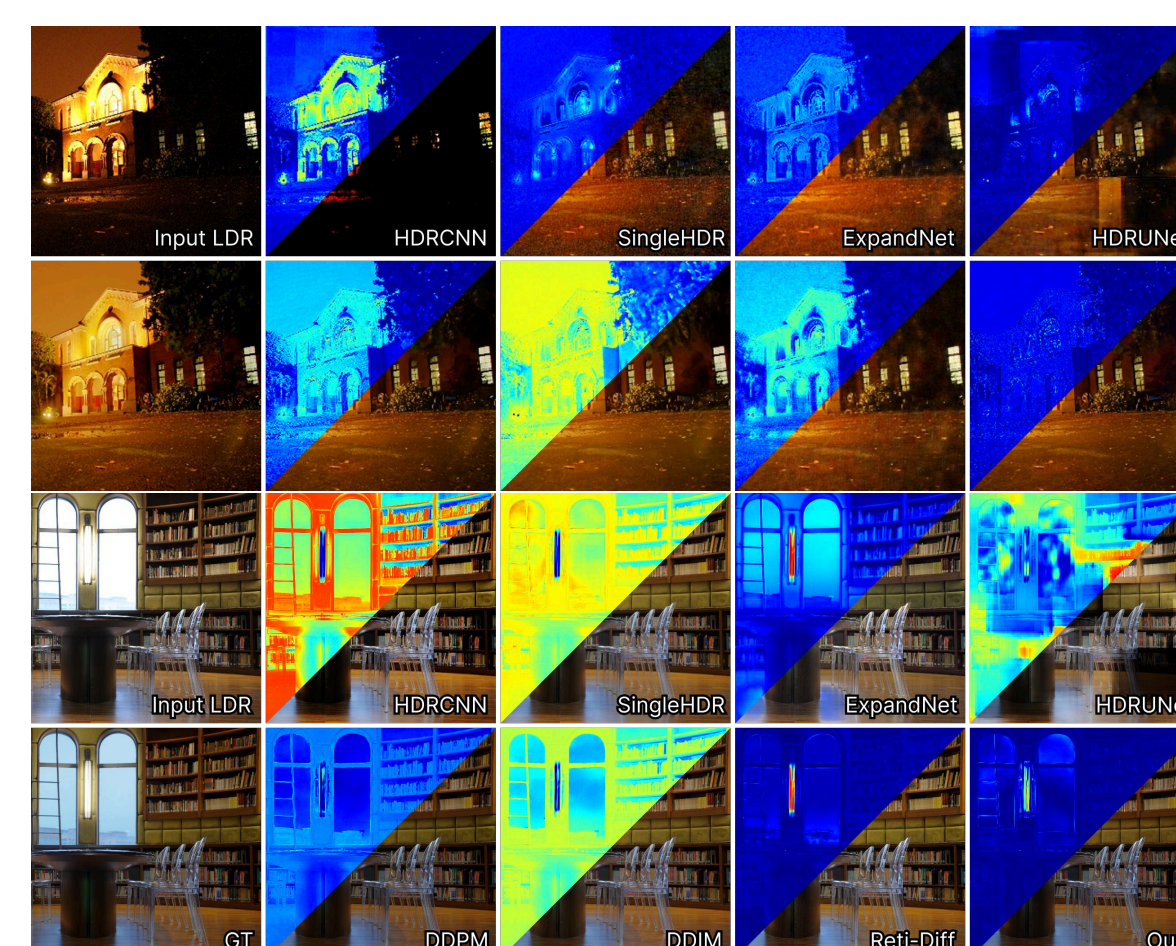
Quantitative Comparison

Dataset	Method	PSNR-μ ↑	SSIM-μ ↑	PSNR-PU ↑	SSIM-PU ↑	PSNR-I ↑	SSIM-I ↑	MS-SSIM ↑	HDR-VDP-2/3 ↑	LPIPS ↓	ΔE2000 ↓
HDR-REAL [24]	HDRCNN [9]	14.99	0.5638	16.25	0.5305	25.81	0.5936	0.7678	31.52/5.71	0.3497	21.14
	SingleHDR [24]	17.92	0.5906	19.93	0.5695	31.08	0.7758	0.8081	34.44/6.16	0.3156	14.36
	ExpandNet [34]	18.07	0.5999	20.21	0.5953	31.70	0.7878	0.8186	30.35/5.91	0.3416	13.77
	HDRUNet [6]	16.22	0.6327	17.79	0.6157	28.35	0.7118	0.7293	24.23/4.41	0.3937	15.72
	DDPM [12]	25.45	0.8173	24.78	0.7901	37.46	0.9063	0.9041	43.52/7.45	0.1921	10.40
	DDIM [45]	20.77	0.6925	22.65	0.6716	34.23	0.8539	0.8307	39.14/6.76	0.2365	14.26
	HDR-Trans. [26]	15.71	0.6027	16.83	0.5532	25.08	0.6239	0.7870	31.72/5.69	0.3665	19.24
	Reti-Diff [11]	27.64	0.8354	29.15	0.8666	35.93	0.9397	0.9147	42.08/7.31	0.2645	4.83
	Ours	28.66	0.8684	30.07	0.8935	36.22	0.9521	0.9304	44.27/7.72	0.1919	4.02
	Ours	28.66	0.8684	30.07	0.8935	36.22	0.9521	0.9304	44.27/7.72	0.1919	4.02
HDR-EYE [36]	HDRCNN [9]	15.55	0.5986	16.12	0.5673	22.84	0.7030	0.8049	37.84/7.08	0.2811	14.05
	SingleHDR [24]	15.04	0.6535	14.36	0.5536	19.04	0.5612	0.8813	45.23/7.66	0.2436	19.28
	ExpandNet [34]	16.09	0.7023	15.15	0.6073	17.05	0.5605	0.8878	27.97/7.32	0.3105	17.55
	HDRUNet [6]	14.81	0.6883	13.99	0.6289	17.69	0.6014	0.8149	26.56/5.79	0.3054	15.35
	DDPM [12]	17.45	0.7296	16.56	0.6859	23.38	0.6191	0.9040	33.12/7.99	0.2005	13.81
	DDIM [45]	16.98	0.7647	16.03	0.7062	21.52	0.6270	0.9044	53.47/7.92	0.2002	13.19
	HDR-Trans. [26]	17.23	0.7453	16.47	0.6889	20.85	0.6576	0.8801	44.62/7.51	0.2537	12.78
	Reti-Diff [11]	15.36	0.6944	14.97	0.6163	18.77	0.5626	0.8974	46.26/7.74	0.2475	17.52
	Ours	20.75	0.8017	19.32	0.7638	21.30	0.7424	0.9053	44.09/7.94	0.2353	9.68
	Ours	20.75	0.8017	19.32	0.7638	21.30	0.7424	0.9053	44.09/7.94	0.2353	9.68
AIM2025 [49]	HDRCNN [9]	17.67	0.6147	16.73	0.6250	23.96	0.6750	0.8097	42.16/6.57	0.3605	10.92
	SingleHDR [24]	20.77	0.7328	21.55	0.7053	28.48	0.7930	0.9027	64.48/7.90	0.2460	10.64
	ExpandNet [34]	24.94	0.8281	25.06	0.8402	29.16	0.8765	0.9455	66.94/8.21	0.2149	5.93
	HDRUNet [6]	25.88	0.8709	26.38	0.8808	30.98	0.9131	0.9312	57.83/7.06	0.2218	4.46
	DDPM [12]	23.03	0.8320	23.36	0.8145	29.71	0.8316	0.9550	75.57/8.78	0.1286	7.91
	DDIM [45]	19.50	0.7733	19.72	0.7277	27.22	0.7602	0.9275	69.96/8.36	0.1580	11.24
	HDR-Trans. [26]	17.12	0.7253	16.70	0.6702	20.26	0.6174	0.8978	62.66/7.78	0.2604	14.94
	Reti-Diff [11]	23.31	0.8269	23.49	0.8229	28.65	0.8370	0.9363	66.93/8.19	0.2332	7.74
	Ours	29.02	0.8922	29.06	0.9069	31.97	0.9360	0.9654	74.01/8.68	0.1511	3.90
	Ours	29.02	0.8922	29.06	0.9069	31.97	0.9360	0.9654	74.01/8.68	0.1511	3.90

References

- HDRUNet Chen et al., CVPR'21
- HDRCNN Eilertsen et al., TOG'17
- Reti-Diff He et al., ICLR'25
- DDPM Ho et al., NeurIPS'20
- HDR-Trans. Liu et al., ECCV'22
- ExpandNet Marnerides et al., CGF'18
- HDR-Eye Nemoto et al., VPQM'15
- DDIM Song et al., ICLR'21

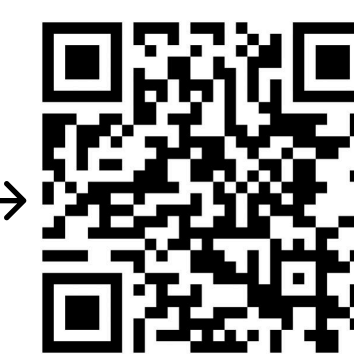
Qualitative Comparison



Conclusion

1. We introduce a one-step generative framework for single-image HDR reconstruction.
2. We design an exposure-aware consistency trajectory for spatially varying LDR degradations.
3. We develop an Exposure-Guided Luminance-Chromaticity (ELC) loss.

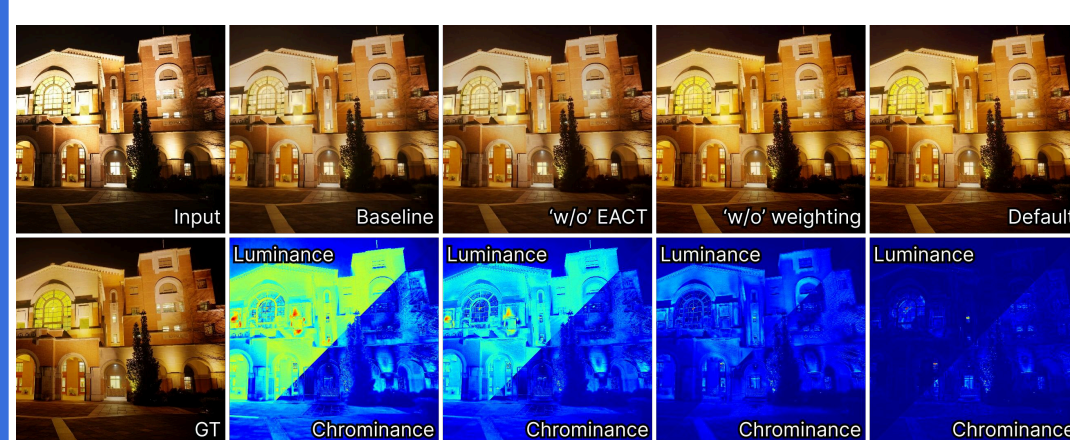
Code & Project



Ablation

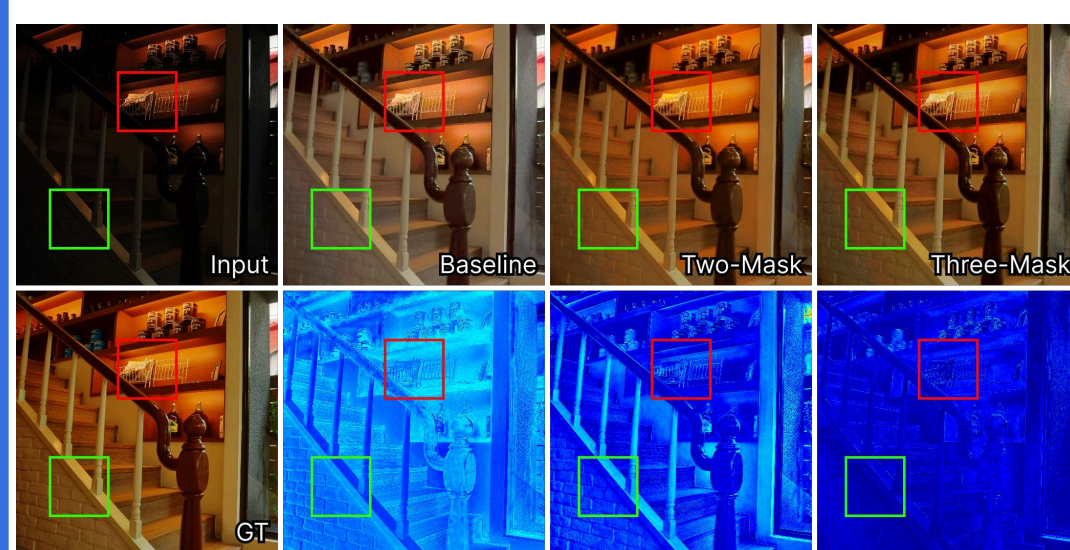
ELC Loss

Method	HDR-REAL [24]			AIM2025 [49]		
	PSNR-μ ↑	SSIM-μ ↑	ΔE2000 ↓	PSNR-μ ↑	SSIM-μ ↑	ΔE2000 ↓
Baseline	21.09	0.6917	12.23	27.90	0.8842	4.24
'w/o' EACT	21.26	0.6988	12.04	28.23	0.8833	4.22
'w/o' weighting	28.56	0.8663	4.06	28.91	0.8898	3.95
Default	28.66	0.8684	4.02	29.02	0.8922	3.90

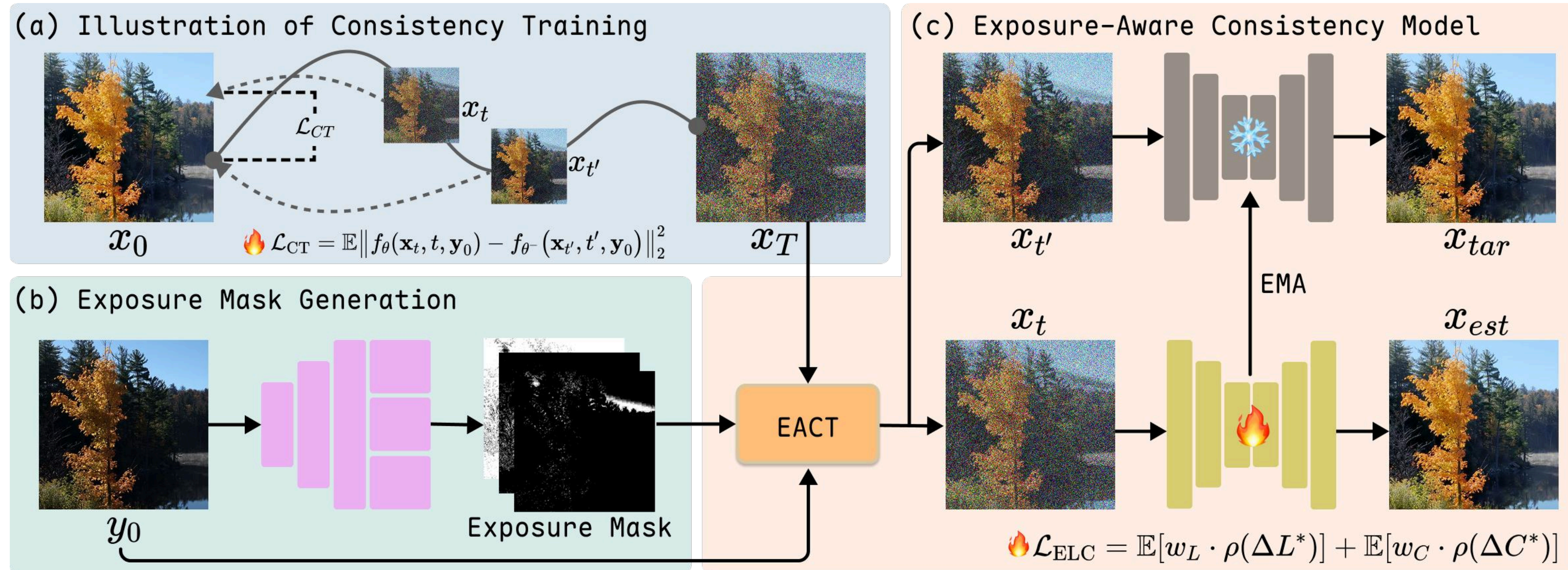


Exposure-Aware Consistency Trajectory

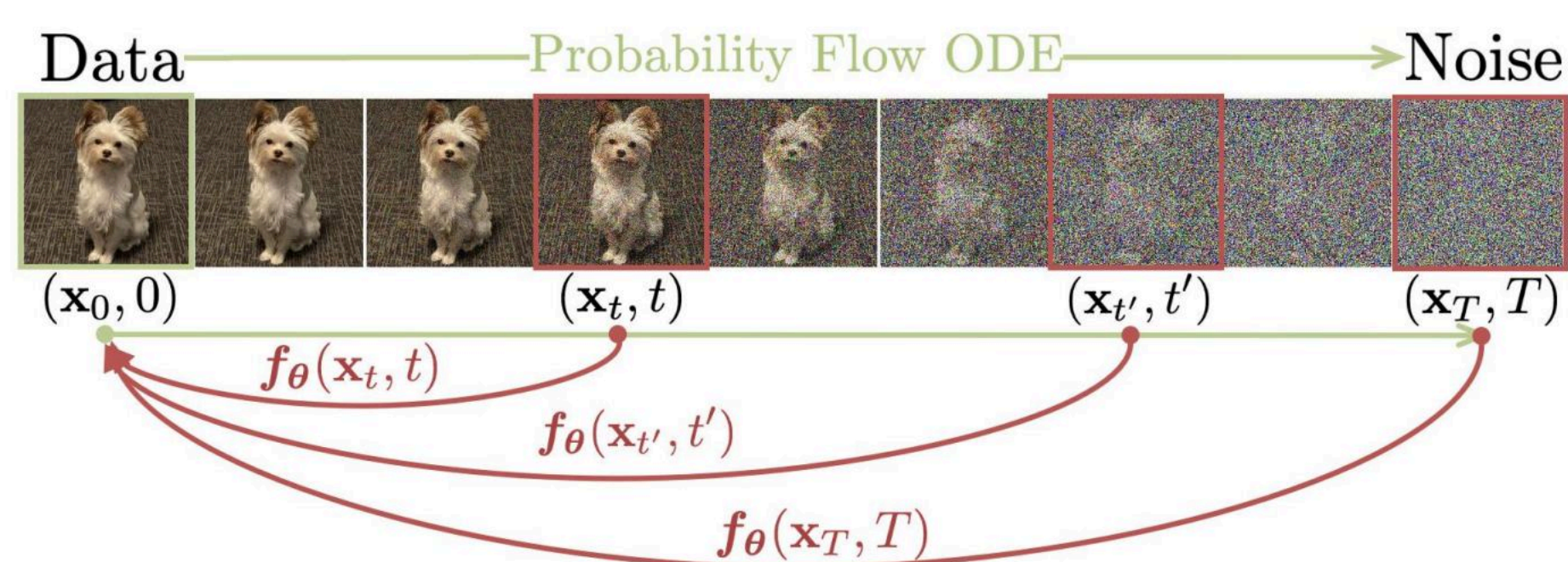
Method	HDR-REAL [24]			AIM2025 [49]		
	PSNR-μ ↑	SSIM-μ ↑	LPIPS ↓	PSNR-μ ↑	SSIM-μ ↑	LPIPS ↓
Baseline	21.09	0.6917	0.3041	27.90	0.8842	0.1920
Two-Mask	25.75	0.8076	0.2785	28.48	0.8936	0.1543
Three-Mask	25.84	0.8282	0.2754	28.89	0.8921	0.1504



ExpoCM: Exposure-Aware One-Step Generative Single-Image HDR Reconstruction



1. One-step generation framework: consistency model



2. Exposure-aware consistency trajectory

3. ELC fintune: Correct chromaticity and luminance deviations